Seif Moheb

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Summary

Computer Science Engineering graduate specializing in AI Engineering and Data Science with practical experience in the end-to-end machine learning lifecycle, from predictive models and OCR pipelines to RAG-based LLM applications. Proven ability to integrate AI solutions into scalable software systems using Python, LangChain, and cloud technologies.

Education

German University in Cairo
Oct 2021 - Jun 2026

Bachelor of Computer Science Engineering - **Grade:** Very Good — *Cairo, Egypt*

Bachelor Thesis: Integrating KRR and CV: The case of YOLO and SNePS - **Grade:** Excellent

Experience

Data Analyst Intern
Aug 2025 – Jan 2026

Enpal — *New Cairo, Egypt*

- Built automated reports using **SQL** and **DAX** in **Power BI** for cross-country teams.
- Created custom queried views using **Snowflake** to speed up semantic model refresh time by **95%**.
- Assisted in onboarding newly hired data analysts with **Snowflake**, **Power BI**, and **SQL**.

AI Engineer Intern
Jul 2024 – Sep 2024

Advansys — *Cairo, Egypt*

- Engineered an **Agentic RAG system** for financial analysis using **LangChain**, **Zephyr-7B**, **Llama-3.1**, and **ChromaDB**, integrating a multilingual **OCR pipeline** (Tesseract, Azure Cognitive OCR) to ingest complex documents.
- Designed a custom **evaluation framework** to assess retrieval quality, achieving **96.5% faithfulness** and **94.6% relevancy** by optimizing embedding strategies (**all-mpnet-base-v2**) and document chunking.
- Created a **Selective Visualization tool** using **Mini-LLMs** and **Plotly** that automatically converts unstructured financial data into analytical charts.

Data Science Intern
Jul 2025 – Aug 2025

Dawar — *Giza, Egypt*

- Prepared ML-ready datasets from rPET and commodity price logs (2020–2025), utilizing **Seaborn** and **Matplotlib** to visualize market correlations and trend anomalies.
- Designed a hybrid price forecasting model by combining time-series regressors with **DistilBERT** for sentiment analysis on socioeconomic indicators.

Projects

- **Graph-RAG FPL Assistant** | *Python, LLMs, RAG, Neo4j, XGBoost, FastAPI, Docker, Azure* — *Dec 2025 – Present*
  - Built an **LLM**-powered **Graph-RAG** assistant for Fantasy Premier League, combining **Cypher** retrieval over a **Neo4j** knowledge graph (**180K+ relationships**) with **semantic search** and **spaCy** entity extraction.
  - Trained an **XGBoost** points model on **166K** player-match rows with a season-based temporal split, improving top-20 pick quality by **11%** over the best baseline (95% bootstrap CI), validated via **ablations** and calibration analysis.
  - Automated an **MLOps** pipeline: daily FPL API ingestion, graph updates, predictions, and **guarded weekly retraining**.
  - Deployed via **Docker Compose** on **Azure** with **GitHub Actions CI/CD** and **200+ automated tests**.
- **X-CV: X-Ray Computer Vision Pipeline** | *Python, React, Vite, PyTorch, YOLO, LLM, CI/CD* — *May 2026*

  Developed an end-to-end computer vision pipeline and **React** web application for automated X-Ray analysis. Integrated multi-stage inference modules for region classification, fracture bounding box extraction utilizing **YOLO, Faster R-CNN, and RTDETR**, and fracture detection. Orchestrated a **Python** backend pipeline with **LLM** reporting capabilities.
- **Code-Switched Arabic QA RAG System** | *Python, PyTorch, Hugging Face, ChromaDB* — *May 2026*

  Architected an end-to-end **RAG** pipeline specialized in processing highly complex, code-switched transcripts mixing **Egyptian dialect**, **Modern Standard Arabic**, and embedded **English**. Engineered a robust text normalization framework and integrated **ChromaDB** to retrieve context and generate answers across dialect-heavy, unstructured data.

Achievements

- Ranked among the **Top 5** teams in Henkel's ThinkTech Hackathon. — *Oct 2025*
- Selected by the Vice-Dean to lead the Computer Engineering Student Curriculum Committee; represented a faculty major of **650 students** in ACQUIN accreditation meetings. — *Mar 2022 - Present*

Technical Skills

Programming Languages: Python, SQL, Java

AI/LLM Tools: PyTorch, TensorFlow, scikit-learn, LangChain, Hugging Face Transformers, ChromaDB, YOLO

Data: NumPy, Pandas, Matplotlib, Seaborn, Plotly, OpenCV

Dev Tools: Git, Jupyter, Docker, OCR (Tesseract)

Databases / Cloud: PostgreSQL, MongoDB, Snowflake, Firebase, Azure, Neo4j

Language Skills

German (C1) — English (C2)